

# Constructing Finite-Order Kernel Models for Performance Evaluation of Kernel Adaptive Filtering in Real-World Scenarios

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**Abstract**—Kernel Adaptive Filters (KAFs) have emerged as powerful tools for nonlinear signal processing. However, a fundamental challenge inherent in KAFs is the linear growth of the dictionary size with the number of input samples, imposing prohibitive computational and memory burdens. While Finite-Order Nonlinear Regression Models offer a viable solution by fixing the model structure, a critical gap remains in the evaluation of robust adaptive algorithms: for real-world nonlinear time series (e.g., sunspot numbers or internet traffic), the “true” system model—comprising an ideal dictionary and optimal weights—is unknown. The absence of ground truth precludes the calculation of key performance metrics, such as the Normalized Mean Square Deviation (NMSD), which is essential for assessing bias compensation techniques. This paper proposes a systematic methodology to construct representative finite-order kernel models directly from nonlinear time series. By employing a coherence-based dictionary selection mechanism and a noise-free pre-training phase, we establish a reliable reference model that serves as the ground truth. We validate this approach on both synthetic benchmarks (Hermite, Lorenz) and real-world datasets (Sunspot, Internet Traffic), demonstrating its utility in quantifying the performance of bias-compensated algorithms in noisy input scenarios.

**Index Terms**—Kernel Adaptive Filter, Finite-Order Model, Nonlinear Regression, System Identification, Time Series Prediction, NMSD.

## I. INTRODUCTION

Nonlinear signal processing is pivotal in modern engineering applications, ranging from channel equalization to chaotic time series prediction. Kernel methods map input data into a high-dimensional Reproducing Kernel Hilbert Space (RKHS), allowing linear algorithms to solve nonlinear problems efficiently. Among these, the Kernel Maximum Correntropy Criterion (KMCC) algorithm has gained attention for its robustness against non-Gaussian, impulsive noise [1].

Despite their theoretical elegance, traditional KAFs suffer from the “curse of dimensionality.” Since the solution is ex-

pressed as a linear combination of kernel evaluations centered at training points, the network size grows indefinitely as processing continues. This uncontrolled growth is impractical for real-time or resource-constrained applications. Consequently, various sparsification techniques, such as the Novelty Criterion (NC) [2]–[4], Prediction Variance Criterion (PVC) [5], Approximate Linear Dependency (ALD) [6], Surprise Criterion (SC) [7], and Quantized Criterion (QC) [8]–[10] have been proposed to curb dictionary growth. However, when dealing with nonlinear systems exhibiting time-varying characteristics, restricting the dictionary size may inadvertently compromise the adaptive capability of the filter. To mitigate this, literature [11] proposed a Re-Labeling mechanism designed to ensure robust tracking performance even under abrupt system changes. Additionally, studies such as [12], [13] have introduced Significance Measurement methods to dynamically evaluate the influence of samples, ensuring that computational complexity remains within manageable limits.

Alternatively, Richard et al. have proposed using Finite-Order Nonlinear Regression Models pre-define a fixed dictionary size, transforming the infinite-dimensional problem into a fixed-size parametric estimation problem [14].

While the finite-order approach effectively bounds the computational complexity, a critical gap remains in the research methodology for robust adaptive filtering studies involving Bias Compensation (BC), specifically regarding performance evaluation. To rigorously evaluate a bias-compensated algorithm, one must compute the Normalized Mean Square Deviation (NMSD):

$$\text{NMSD} = 10 \log_{10} \frac{\|\omega - \hat{\omega}(n)\|^2}{\|\omega\|^2}, \quad (1)$$

where  $\omega$  and  $\hat{\omega}(n)$  denote the optimal and estimated at time  $n$  weight vector, respectively. In synthetic experiments,  $\omega$  is known by design. However, for real-world phenomena like chaotic weather patterns or network traffic, neither the optimal dictionary nor the true weights exist a priori. Without a

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ground truth, researchers are limited to using Mean Squared Error (MSE), which combines system noise with estimation error, thereby obscuring the true convergence behavior of the algorithm.

This paper bridges this gap by proposing a rigorous framework to “reverse engineer” a finite-order kernel model from observed time series. By extracting a diverse dictionary and identifying optimal weights from clean data segments, we create a proxy ground truth. This enables the precise evaluation of advanced algorithms using NMSD on real-world datasets.

## II. METHODOLOGY: MODEL CONSTRUCTION

We consider a nonlinear system output  $y(n)$  approximated by a fixed dictionary  $\mathcal{D} = \{\mathbf{d}_1, \dots, \mathbf{d}_M\} \in \mathbb{R}^{L \times M}$  of size  $M$ :

$$y(n) = \sum_{j=1}^M \omega_j \kappa(\mathbf{u}(n), \mathbf{d}_j) + v(n) \quad (2)$$

where  $\mathbf{u}(n) \in \mathbb{R}^{L \times 1}$  is the input vector with embedded time  $L$ ;  $\kappa(\cdot, \cdot)$  is the Gaussian kernel with bandwidth  $\sigma$ , and  $\omega_j$  is the  $j$ -th weight of the optimal weight vector  $\boldsymbol{\omega} \in \mathbb{R}^{M \times 1}$ . The purpose of model construction is to obtain a data-driven weight vector  $\boldsymbol{\omega}$  and dictionary  $\mathcal{D}$  for the nonlinear regression model.

### A. Coherence-Constrained Random Dictionary Generation

Constructing an efficient finite-order model requires the dictionary  $\mathcal{D} = \{\mathbf{d}_1, \dots, \mathbf{d}_M\}$  to be both representative and non-redundant. While we adopt a data-independent strategy by generating dictionary atoms from a Gaussian distribution  $\mathcal{N}(\hat{\mu}_u, \hat{\sigma}_u^2 \mathbf{I})$  derived from the input statistics, pure random sampling can lead to atom clustering, wasting the limited model budget  $M$ .

To mitigate this, we incorporate a coherence-based rejection mechanism during the initialization phase. A randomly generated candidate atom  $\mathbf{d}_{cand}$  is accepted into  $\mathcal{D}$  only if it is sufficiently distinct from the already selected atoms. The coherence check is defined as:

$$\max_{d_j \in \mathcal{D}} |\kappa(\mathbf{d}_{cand}, \mathbf{d}_j)| < \delta, \quad (3)$$

where  $\delta \in [0, 1]$  is a pre-defined coherence threshold. If the condition is met,  $\mathbf{d}_{cand}$  is added to  $\mathcal{D}$ ; otherwise, it is discarded, and a new candidate is regenerated. This ensures that the mutual coherence of the dictionary remains low, forcing the random centers to span the input space in a diverse manner. Note that even though the conventional K-Means method can provide representative centers, it requires iterative training on the dataset [15]. Our coherence-constrained random approach provides a comparable spatial coverage with significantly lower computational complexity (one-shot generation) and does not require storing the full training dataset during initialization.

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### Algorithm 1 Random Dictionary Construction with Coherence Constraint

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**Require:** Input stats  $\hat{\mu}_u, \hat{\sigma}_u$ , Order  $M$ , Threshold  $\delta$

**Ensure:** Fixed Dictionary  $\mathcal{D}$

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1: Initialization:
2: Generate first atom:  $\mathbf{d}_1 \leftarrow \text{Randn}(\hat{\mu}_u, \hat{\sigma}_u)$ 
3:  $\mathcal{D} \leftarrow \{\mathbf{d}_1\}$ 
4: Dictionary Growth:
5: while  $|\mathcal{D}| < M$  do
6:   Candidate Generation:
7:    $\mathbf{d}_{cand} \leftarrow \text{Randn}(\hat{\mu}_u, \hat{\sigma}_u)$ 
8:   Coherence Check:
9:   Compute max coherence:  $\mu = \max_{d \in \mathcal{D}} |\kappa(\mathbf{d}_{cand}, \mathbf{d})|$ 
10:  Inclusion Test:
11:  if  $\mu < \delta$  then
12:    Accept candidate:  $\mathcal{D} \leftarrow \mathcal{D} \cup \{\mathbf{d}_{cand}\}$ 
13:  else
14:    Discard  $\mathbf{d}_{cand}$  (Redundant)
15:  end if
16: end while
17: return  $\mathcal{D}$ 

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### B. Optimal Weight Acquisition

Once the dictionary  $\mathcal{D}$  is fixed, the ‘true’ weights  $\boldsymbol{\omega}$  are determined via a numerical identification approach, chosen to circumvent the complexity of deriving analytical solutions for MCC-based costs. Specifically, the process begins with noise-free training using a clean segment of the time series devoid of measurement noise. A standard KAF algorithm, such as KMCC, is then executed on this data until steady-state convergence is achieved. Finally, the converged weight vector is extracted, frozen, and designated as the optimal weight vector  $\boldsymbol{\omega}$ . This derived model  $(\mathcal{D}, \boldsymbol{\omega})$  becomes the benchmark for all subsequent noisy experiments.

### C. Model Selection and Validation

Since the dictionary centers are generated stochastically, the modeling performance may vary depending on the initialization. To ensure the constructed finite-order model faithfully represents the underlying nonlinear dynamics (and serves as a reliable ground truth for NMSD evaluation), we employ a *Monte Carlo validation strategy*.

For each dataset, we conduct  $K$  independent trials (e.g.,  $K = 100$ ) of the proposed dictionary generation and weight identification process. Let  $\{\mathcal{D}_i, \boldsymbol{\omega}_i\}$  denote the model obtained in the  $i$ -th trial. We evaluate the fidelity of each model using the MSE on a separate validation set  $\mathcal{V}_{val}$ :

$$\text{MSE}_i = \frac{1}{|\mathcal{V}_{val}|} \sum_{n \in \mathcal{V}_{val}} (y(n) - \hat{y}_i(n))^2 \quad (4)$$

The final model  $(\mathcal{D}^*, \boldsymbol{\omega}^*)$  is selected as the one yielding the minimum validation error:

$$(\mathcal{D}^*, \boldsymbol{\omega}^*) = \arg \min_{(\mathcal{D}_i, \boldsymbol{\omega}_i), i=1 \dots K} \text{MSE}_i \quad (5)$$

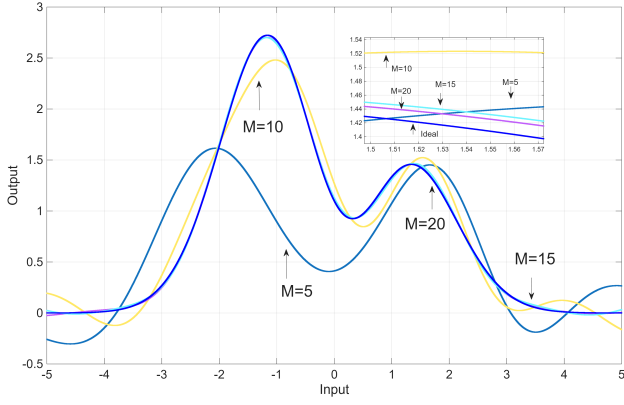


Fig. 1. Evaluation of impact of  $M$  on the synthesized results for Hermite function

This best-performing model is then fixed and used as the reference system for all subsequent adaptive filtering experiments.

### III. EXPERIMENTAL CASE STUDIES

We validated the proposed framework on four datasets ranging from mathematical functions to real-world traffic data. In all cases, we demonstrate how the constructed model allows for the evaluation of the Bias-Compensated KMCC (BC-KMCC) algorithm against the standard KMCC.

#### A. Synthetic Benchmark: Hermite Function

The Hermite function [16] is a standard nonlinearity defined as:

$$y(n) = 1.1 [1 - u(n) + 2u^2(n)] \exp\left(-\frac{u^2(n)}{2}\right) \quad (6)$$

where we evaluate the input  $u(n)$  that is uniformly distributed in  $[-5, 5]$ . As shown in Fig. 1, we evaluate the impact of the number of dictionaries  $M$  on the synthesized results with kernel bandwidth  $\sigma = 1.0$ . The coherence threshold  $\delta = 0.95$  in this example ensures that we can collect enough dictionary, which is suggested as  $M = 20$ .

#### B. Chaotic Dynamics: Lorenz Time Series

The Lorenz system is described by the differential equations:

$$\frac{dx}{dt} = \zeta(y - x), \quad \frac{dy}{dt} = x(\iota - z) - y, \quad \frac{dz}{dt} = xy - \theta z \quad (7)$$

with parameters  $\zeta = 10$ ,  $\iota = 28$ , and  $\theta = 8/3$ . We choose the  $x$ -component time series as an example to see how to construct the model. Note that data was normalized to  $[0, 1]$  using Eq. (8):

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (8)$$

The input vector dimensionality was  $L = 5$ , i.e.,  $\mathbf{u}(n) = [x(n), \dots, x(n-4)]^T$ .

With a dictionary of  $M = 10$  selected from the chaotic attractor, we derived the optimal weights. Fig. 2 demonstrates

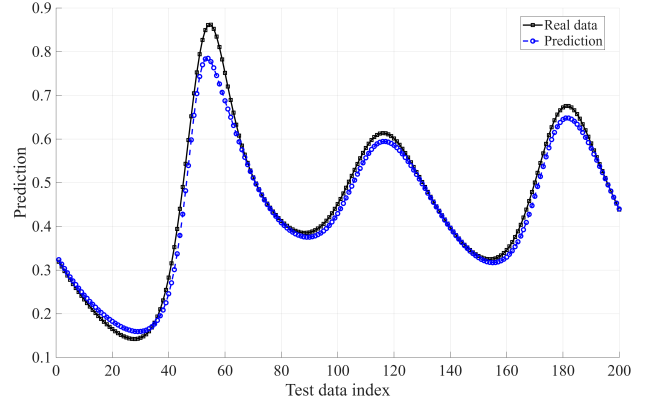


Fig. 2. Experimental results on the Lorenz Time Series

the tracking capability of the constructed model ( $M = 10$ ) on the Lorenz chaotic time series. While the model accurately synchronizes with the phase and frequency of the system, noticeable deviations are observed at the local maxima (e.g., around indices  $n = 55$  and  $n = 180$ ). These discrepancies are attributed to the limited dictionary size constrained by the coherence threshold. Since the Lorenz attractor exhibits high curvature at its extremities (turning points), a sparse distribution of Gaussian kernels may result in a smoothed approximation at these peaks. However, the overall dynamics are preserved, validating the model's utility as a reference system.

#### C. Real-World Data: Sunspot Numbers

We utilized the monthly mean total sunspot number dataset provided by SIDAC, covering the period from January 1749 to January 2025 (3313 samples)<sup>1</sup>. Sunspot activity correlates with solar flares and geomagnetic storms, making prediction critical.

The non-stationary nature of sunspot data makes modeling challenging. We used an input dimension of  $L = 3$  and  $M = 10$ . Normalization was applied to map the count data to  $[0, 1]$ . The coherence threshold was set to 0.9, resulting in a sparse dictionary selection.

Fig. 3 illustrates the prediction performance of the proposed finite-order model on the Sunspot time series. Despite the inherent non-stationary and chaotic nature of the dataset, the constructed model demonstrates a strong capability to track the complex dynamics of the target signal. It is observed that the predicted trajectory (blue dashed line) maintains a high degree of congruence with the real data (black solid line) across the test samples. This validates that the selected dictionary size of  $M = 10$ , derived via the coherence-constrained mechanism, is sufficient to capture the underlying system behavior without overfitting.

#### D. Real-World Data: Internet Traffic

Finally, we analyzed the A5M Time Series dataset, representing internet traffic data from a European ISP collected

<sup>1</sup><https://www.sidc.be/>

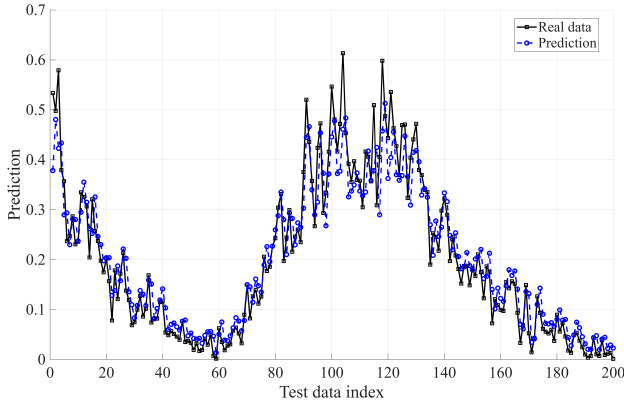


Fig. 3. Experimental results on the Sunspot dataset (1749-2025).

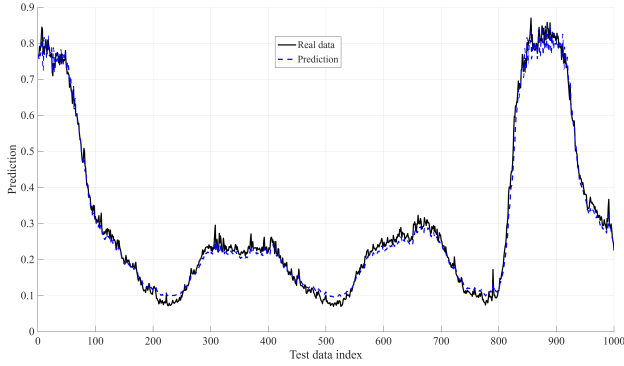


Fig. 4. Experimental results on the Internet Traffic dataset.

over nearly two months<sup>2</sup>. Network traffic exhibits strong daily periodicity but is prone to bursty anomalies.

We selected 13,000 samples for training and 1,000 for testing, with  $L = 5$  and  $M = 10$ . The coherence check confirmed that the selected dictionary atoms represented distinct traffic loads (peak vs. off-peak). Fig. 4 depicts the prediction performance on the real-world Internet Traffic dataset. Network traffic is characterized by strong daily periodicity and bursty anomalies, which pose significant challenges for modeling and analysis. However, the proposed finite-order model effectively tracks these rapid fluctuations, as evidenced by the tight alignment between the predicted curve (dashed line) and the actual traffic load (solid line). This demonstrates the model's robustness in handling non-stationary variations, which are typical of real-world ISP networks.

#### IV. DISCUSSION

It was observed that due to the localized nature of Gaussian kernels, random centers failing to cover the high-curvature regions of the chaotic attractor resulted in higher modeling errors. However, the proposed coherence-constrained sampling, combined with the validation-based selection strategy, effectively mitigated this issue, consistently yielding high-precision models with Normalized MSEs below a desired accuracy of modeling.

<sup>2</sup><https://pcortez.dsi.uminho.pt/downloads>

#### V. CONCLUSION

This paper addressed the dual challenge of dictionary growth and the lack of ground truth in nonlinear adaptive filtering. We presented a systematic workflow to build finite-order nonlinear regression models using kernel functions. By leveraging coherence-based dictionary selection and noise-free weight identification, we constructed reliable reference models for both synthetic and real-world time series.

Experimental validation on Hermite functions, Lorenz attractors, sunspot numbers, and internet traffic data confirmed that these models are not only computationally efficient but also essential for the rigorous evaluation of robust algorithms. The proposed framework enables researchers to utilize NMSD as a performance metric in practical scenarios, facilitating the development of more effective bias compensation techniques for real-world applications.

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